



Detect the Obstacles
around you !

RP Lidar & Benewake

- ◆ *Field of view of 13° to 360°*
- ◆ *Detection range 6m to 180m*
- ◆ *Infrared 3D Lidar to solid state Lidar*



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COMPARISON BETWEEN DIFFERENT MACHINE LEARNING API LIBRARIES

Feature	H2O	SparkMLlib	Sparkling Water	Weka
Open source/closed source	Open	Open	Open	Open, but licensed under GPL 2.0 for commercial use
Algorithms available	Deep learning, generalised linear modelling, gradient boosting machine, K-Means, standard statistics like min, max, mean, stddev	Classification, regression, clustering, collaborative filtering, dimensionality reduction, optimisation primitives	Deep learning, generalised linear modelling, gradient boosting machine, K-Means, standard statistics like min, max, mean, stddev	Classifiers, clusterers, associations, attribute selection, preprocessing filters
Languages supported	R, Phtyon, Scala, Java	Python, Scala, Java	R, Python, Scala, Java	Java, Octave/Matlab, R, Python
Efficiency with large datasets	Moderate	Good	Good	Moderate
Scalable	Yes	Yes	Yes	No
Community support	Good	Good	Good	Good

cial API library in Spark engine. It supports classification, logistic regression, Naïve Bayes, Random Forest, K-Means and Gaussian mixtures ML algorithms. This library can be used in Python or Scala programming language.

To know whether ML can be applied for the Internet of Things (IoT) processing, we have to understand abnormal situations that can occur in IoT networks. On application layer of the IoT, broken sensors might send wrong sensed values to a connected server continuously. If the intelligent network is not aware of this

problem, it will not be able to detect accidents in IoT networks.

ML can be applied on sensor networks. ISM band is used by 802.11, 802.15.4 and 802.15.1 wireless protocols. When different devices are scattered in one area, there is a different system interference. ML can be used to classify wireless channel errors in different categories, and this can be used to diagnose problems in IoT networks.

In wireless sensor networks, fault detection is a complicated process. Faults caused by degraded or malfunctioning

sensors are classified as data faults. Fault types caused by low battery, calibration, communication or connection failures are classified as system faults. When ML is applied, faults—such as gain, out-of-bound, offset and stuck-at—can be recognised in detail. Classifying faults can also help in IoT networks.

It is not easy to design fault-tolerant systems in constrained IoT networks. People are working on designing IoT systems that can detect, predict and recover from abnormal situations. ML is helping improve IoT networks, too. **EFY**